EMPLOYEE ATTRITION & WORKPLACE ANALYTICS

Exploratory Data Analysis Report

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Dataset: IBM HR Analytics Employee Attrition & Performance

Project Objective

To explore employee attrition patterns and examine how selected employee, job and workplace characteristics are associated with attrition.

Dataset note: This is a fictional/synthetic dataset and does not represent actual IBM employee records.

1. Executive Summary

The dataset contains 1,470 employee records. The original dataset has 35 columns. After removing three constant columns and the employee identifier, 31 columns remained for analysis. Overall attrition was 16.12% (237 employees); 83.88% (1,233 employees) did not leave.

Measure	Result
Employee records	1,470
Original columns	35
Columns after cleaning	31
Attrition = Yes	237 (16.12%)
Attrition = No	1,233 (83.88%)
Missing values	None found
Duplicate rows	None found

2. Dataset and Tools

The analysis uses the IBM HR Analytics Employee Attrition & Performance CSV dataset. Python was used in Jupyter Notebook with Pandas and NumPy for data handling, and Matplotlib and Seaborn for visualization.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn and Jupyter Notebook.

3. Data Cleaning

- Inspected the dataset structure, column names, data types and descriptive statistics.
- Checked for missing values; none were reported.
- Checked for duplicate rows; none were found.
- Identified EmployeeCount, Over18 and StandardHours as constant columns.
- Removed those three constant columns and EmployeeNumber, an identifier column.
- The cleaned analysis dataframe contained 1,470 rows and 31 columns.

4. Exploratory Data Analysis: Key Findings

Overtime: Attrition was 30.53% among employees working overtime, compared with 10.44% among employees not working overtime.

Department: Sales had the highest department-level attrition rate (20.63%), followed by HR (19.05%) and Research & Development (13.84%).

Job role: Sales Representatives had the highest job-role attrition rate (39.76%). Laboratory Technicians (23.94%) and Human Resources employees (23.08%) also showed comparatively higher rates.

Age: The 18–25 age group had an observed attrition rate of 34.78% in the grouped analysis.

Marital status: Single employees had an attrition rate of 25.53%, compared with 12.48% for married employees and 10.09% for divorced employees.

Job satisfaction: The lowest satisfaction level had a 22.84% attrition rate, compared with 11.33% at the highest level.

Work-life balance: The lowest work-life-balance level had a 31.25% attrition rate.

Business travel: Frequently travelling employees had a 24.91% attrition rate, compared with 14.96% for rare travel and 8.00% for non-travel.

Tenure: The grouped 0–1 year tenure category had a 34.88% attrition rate. The 2–5, 6–10 and 11–20 year groups showed 15.51%, 12.28% and 6.67%, respectively.

Job involvement: The lowest job-involvement level had a 33.73% attrition rate, compared with 9.03% at the highest level.

Environment satisfaction: The lowest environment-satisfaction level had a 25.35% attrition rate, compared with 13.45% at the highest level.

Distance from home: The grouped 20–30 km category showed a 22.06% attrition rate, while the 0–5 km category showed 13.77%.

5. Visualizations

The notebook includes three histograms, three box plots, bar charts, two count plots, two scatter plots, one correlation heatmap and two additional visualizations. A time-series plot was not considered applicable because the dataset does not contain a suitable date/time variable.

Histograms show numerical distributions; box plots help inspect spread and potential outliers; count plots and bar charts compare categories and attrition patterns; scatter plots explore relationships between numerical variables; and the heatmap summarizes pairwise numerical correlations.

6. Recommendations

- Review overtime patterns and workload distribution.
- Investigate employee experience in departments and job roles with higher observed attrition, particularly Sales and Sales Representative roles.
- Strengthen onboarding, mentoring and early-career support for employees with shorter tenure.
- Monitor employee satisfaction, work-life balance and job involvement through regular feedback.
- Use further investigation and employee feedback before making policy changes; descriptive patterns alone do not establish causes.

7. Conclusion and Limitations

This exploratory analysis highlights differences in attrition rates across employee and workplace groups. Overtime, department, job role, tenure, satisfaction measures and business travel show useful patterns for further investigation. These results can guide HR questions, but they do not prove that any individual factor causes an employee to leave.

Limitations: The dataset is fictional and may not represent a real organization. The analysis is descriptive, not causal. Some categories may contain relatively few employees, so their percentages can be unstable. Grouped numerical results depend on the selected bin boundaries. Check the final notebook to ensure every reported figure matches the executed cells.